

Improving Patient-Ventilator Synchrony During Pressure Support Ventilation Based on Reinforcement Learning Algorithm

Liming Hao , Xiaohan Wang, Shuai Ren , *Member, IEEE*, Yan Shi , Maolin Cai, Tao Wang ,
and Zujin Luo 

I. INTRODUCTION

Abstract—Mechanical ventilation is an effective treatment for critically ill patients and those with pulmonary diseases. However, patient-ventilator asynchrony (PVA) remains a significant challenge, potentially leading to high mortality. Improving patient-ventilator synchrony poses a complex decision-making problem in clinical practice. Traditional methods rely heavily on clinicians' experience, often resulting in inefficiencies, delayed ventilator adjustments, and resource shortages. This paper proposes a novel approach using a deep reinforcement learning (RL) algorithm based on deep Q-learning (DQN) to enhance patient-ventilator synchrony during pressure support ventilation. The action space and reward function are established from clinical experience, and a pneumatic model of the mechanical ventilation system is constructed to simulate various patient conditions and types of PVAs. Clinical data are used to evaluate the RL algorithm qualitatively and quantitatively. The RL-optimized ventilation strategy reduces the proportion of breaths containing PVAs from 37.52% to 7.08%, demonstrating its effectiveness in assisting clinical decision-making, improving synchrony, and enabling intelligent ventilator control, bedside monitoring, and automatic weaning.

Index Terms—Patient-ventilator synchrony, deep reinforcement learning, decision-making optimization, pressure support ventilation.

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Liming Hao is with the Systems Engineering Research Institute, China State Shipbuilding Company Ltd., Beijing 100097, China.

Xiaohan Wang, Shuai Ren, and Tao Wang are with the School of Automation, Beijing Institute of Technology, Beijing 100081, China (e-mail: renshuai@bit.edu.cn).

Yan Shi and Maolin Cai are with the School of Automation Science and Electrical Engineering, Beihang University, Beijing 100191, China.

Zujin Luo is with the Department of Respiratory and Critical Care Medicine, Beijing Engineering Research Center of Respiratory and Critical Care Medicine, Beijing Institute of Respiratory Medicine, Beijing Chao-Yang Hospital, Capital Medical University Beijing, Beijing 100020, China (e-mail: xmjg2002@mail.ccmu.edu.cn).

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MECHANICAL ventilation is an essential treatment for patients with pulmonary diseases or critical illnesses. During the treatment, patient-ventilator asynchronies (PVAs) occur when the ventilator cannot match the patient's breathing effort. PVAs affect around 25% of patients with acute respiratory failure treated by invasive mechanical ventilation [1], [2]. And synchrony problems occur in up to 80% of patients treated with non-invasive ventilation [3]. According to the description in [4], more than 10% of patients in weaning centers suffer from PVAs. These asynchronies are associated with prolonged ventilation time, respiratory complications, and high mortality rates.

Considering the harmful effects of PVAs, improving patient-ventilator synchrony has become a critical research topic in the field of respiratory medicine. The challenge lies in the complexity of clinical decision-making during mechanical ventilation, as patient conditions vary significantly depending on diseases and fluctuate dynamically during treatment. Adjusting ventilator settings requires substantial clinical expertise and continuous monitoring, which places a heavy burden on clinicians.

This article proposes a deep reinforcement learning (RL) algorithm to improve patient-ventilator synchrony during pressure support ventilation. By modeling the ventilator as the agent in the RL framework and the mechanical ventilation system as the environment, the interaction between the patient and ventilator is treated as states, and actions are performed by adjusting the ventilator's control settings. The proposed approach leverages deep Q-networks (DQN) to handle the complex decision-making process during mechanical ventilation. Both clinical and simulated data are utilized to validate the effectiveness of the proposed algorithm through ablation studies.

II. RELATED WORK

During mechanical ventilation, clinicians adjust the controlling parameters of the ventilator according to clinical experience and medical knowledge to reduce PVAs and improve patient-ventilator synchrony. In [5], Robert L. Chatburn et al. described the types of patient-ventilator asynchronies and outlined the potential types of patient-ventilator interactions based on the phases of breathing. In [6], L. Hao et al. proposed a novel method to evaluate patient-ventilator synchrony. In [4], Arnaud

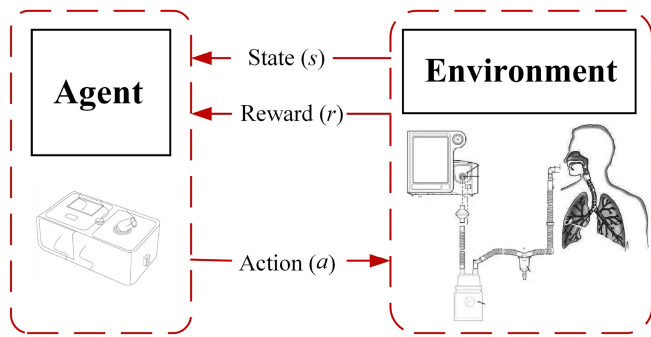


Fig. 1. Process schematic of Reinforcement Learning applied on the mechanical ventilation.

W. Thille et al. summarized clinical methods for promoting patient-ventilator synchrony. According to their research, synchrony can be optimized during pressure support ventilation by adjusting parameters such as the inspiratory trigger level, positive end-expiratory pressure (PEEP), pressure support level, and inspiratory time [7].

As ventilation techniques develop, novel ventilatory modes such as proportional assist ventilation (PAV), neurally adjusted ventilatory assist (NAVA), and adaptive support ventilation (ASV) have been introduced to promote patient-ventilator synchrony [8]. However, each ventilation mode has its own advantages and limitations. Adjustments to ventilator control parameters and transitions among different modes still rely heavily on clinicians' experience. Moreover, timely treatment for PVAs requires continuous monitoring of the patient's ventilation conditions, which demands significant time and effort from clinicians.

The development of artificial intelligence (AI) provides new methods to address PVA-related issues [36]–[38]. Among these, reinforcement learning (RL), a specific area of machine learning, has recently been used to optimize decision-making and treatment policies during mechanical ventilation and other medical applications [35]. Since Sutton et al. proposed the reinforcement learning algorithm [9], RL has demonstrated significant advantages in solving complex sequential decision-making problems.

The general framework for applying RL to mechanical ventilation is illustrated in Fig. 1, where the ventilator is modeled as the agent and the mechanical ventilation system is modeled as the environment. Patient conditions are treated as states, and ventilator control parameters, such as pressure support level or inspiratory trigger level, are treated as actions. This framework highlights the interaction between the patient and the ventilator, which forms the basis for RL-based optimization.

In healthcare, RL has been applied to tasks such as automated radiation adaptation in lung cancer [10], seizure control [11], optimizing antiretroviral therapy for HIV [12], sepsis management [13], and treatments for diabetes [14], anemia [15], anesthesia [16], schizophrenia [17], and other diseases [18]. These studies demonstrate RL's potential to optimize treatment strategies in complex medical environments.

Complex clinical decision-making also affects treatment outcomes during mechanical ventilation [39]. Respiratory parameters vary greatly across pulmonary diseases such as chronic obstructive pulmonary disease (COPD), acute respiratory distress syndrome (ARDS), obesity, idiopathic fibrosis, and others [30], [40]. Different ventilation strategies are adopted based on the disease, and patient conditions often fluctuate dynamically during mechanical ventilation. These fluctuations may lead to conflicts between the patient and the ventilator, necessitating real-time adjustments to ventilator settings, such as triggers, switching from inspiration to expiration, pressure support levels, respiratory rates, and ventilation modes. These patient-ventilator interactions pose complex decision-making challenges.

Several RL-based algorithms have been proposed to address such challenges. Niranjani Prasad et al. [21] developed an RL-based approach for mechanical ventilation weaning in ICUs, applying off-policy RL algorithms to determine optimal actions from ICU data. Y. Chao et al. [22] introduced a supervised-actor-critic (SAC) algorithm for intelligent mechanical ventilation and sedative dosing. Arne Peine et al. [23] developed and validated an RL algorithm to dynamically optimize mechanical ventilation for critically ill patients. The optimized ventilator settings in their study included PEEP, fraction of inspired oxygen (FiO_2), and tidal volume (V_t). These studies primarily focused on optimizing the weaning process or improving outcomes such as reintubation rates, hospital stay durations, and patient mortality. However, patient-ventilator synchrony was not specifically considered as an optimization objective.

Sufficient data and the construction of the environment are prerequisites for training RL algorithms. Previous studies on applying RL to mechanical ventilation relied on clinical or open databases, which often lack specific annotations for PVAs. A. van Diepen et al. [24] proposed a model-based approach to generate annotated pressure support waveforms simulating different types of PVAs. T.H.G.F. Bakkes et al. [25] developed a machine learning method for automatic detection and classification of PVAs based on annotated clinical data. Inspired by their work, we developed a pneumatic model of the mechanical ventilation system to simulate ventilation conditions at various synchrony levels and created our own clinical dataset for analyzing and classifying PVAs.

In contrast to previous works, this study focuses on improving patient-ventilator synchrony during pressure support ventilation, with synchrony as the primary optimization goal. Traditional methods often rely on clinical expertise or simplified models, which may fail to capture the complex dynamics of patient-ventilator interactions. We propose a RL algorithm based on DQN, which uses clinical and simulated data to optimize ventilator parameters and reduce asynchrony. The discrete nature of ventilator adjustments makes DQN particularly well-suited for this task. As a model-free RL approach, DQN does not require an explicit system model, enabling it to address the complex optimization problem in mechanical ventilation effectively. This work demonstrates the potential of the DQN-based deep learning algorithm in improving patient-ventilator synchrony and lays the foundation for further refinement of this approach.

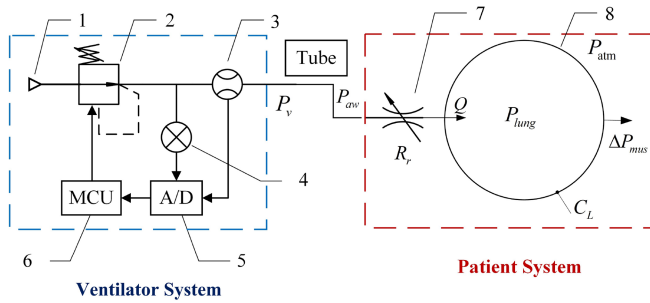


Fig. 2. The schematic of the mechanical ventilation system. 1, air supply resource, 2, control valve, 3, flow sensor, 4, pressure sensor, 5, data acquisition module, 6, control center, 7, throttle valve, 8, active lung simulator.

III. MATERIALS AND METHODS

A. Clinical Dataset and Preprocessing

The clinical dataset used in this study was collected from Beijing Chao-Yang Hospital with approval from the Ethics Committee (Ethical approval number: 2022-KE-412). This dataset includes respiratory waveforms such as pressure, flow, and tidal volume, recorded from 180 patients aged 42–65 years (40% female, 60% male). The data were collected under pressure support ventilation (PSV) and covered common patient-ventilator asynchronies (PVAs), including ineffective triggering, delayed triggering, premature cycling, delayed cycling, and double triggering. A rigorous preprocessing pipeline was applied to ensure data quality. Abnormal signals caused by sensor disconnection, patient movement, and ventilation mode switching were identified and removed based on clinically defined thresholds: pressure values were constrained to 0–25 cmH₂O, flow to –100 to 100 L/min, and tidal volume to 0–1000 mL. Signals were resampled to a uniform frequency of 100 Hz, and z-score normalization was applied for standardization.

B. Pneumatic Model of the Mechanical Ventilation System

The control strategies of the ventilator are simulated based on the pneumatic model of the mechanical ventilation system. In our former research [6], [19], [34], we have proposed a pneumatic model of the mechanical ventilation system. The model has considered the spontaneous breaths of the patient. The adopted ventilation is pressure support ventilation. The schematic of the dynamic model is shown as Fig. 2.

The patient-ventilator synchrony is evaluated according to the interaction process between the patient and ventilator. The patient and ventilator systems are regarded as two pneumatic systems. In this paper, we focus on promoting the patient-ventilator synchrony under the pressure support ventilation. The patient system is driven by the effort from respiratory muscles. The ventilator is triggered when the patient's effort reaches the trigger level and switches from inspiration to expiration when the flow reaches to the cycling level. The patient-ventilator asynchrony occurs when the ventilator cannot match the patient's breathing timestamps.

The pneumatic model is developed from the research in [6] and [19]. In the mechanical ventilation system, the mathematical relationship among the pressure, flow and volume is constructed based on the ideal gas state equation ($PV = mRT$) [20]. As shown in Fig. 2, the pulmonary pressure of the patient is represented as P_{lung} while the output pressure of the ventilator is represented as P_v . The flow (Q) is generated from the pressure difference between the P_{lung} and P_v . The tidal volume (V) is equal to the integration of volumetric flow. The equations are shown as the follows.

1) Pressure equation

$$\frac{dP_{lung}}{dt} = \frac{\rho RT Q V}{V^2 + C_L m R T} \quad (1)$$

2) Flow equation

$$Q = \frac{n A_e P_u \sqrt{1-b}}{\rho \sqrt{R T}} \sqrt{1 - \left(\frac{P_d - b}{1-b} \right)^2} \quad (2)$$

3) Volume equation

$$dV = C_L \cdot dP_{lung} \quad (3)$$

As shown in (1), the air density (1.293 kg/m³) is represented as ρ , the gas constant (297 J/(kg·K)) is represented as R . T is the temperature, Q is the volumetric flow rate, V is the lung volume, C_L is the pulmonary compliance, and m is the air mass. In the pneumatic system, the airway resistance of the patient is simulated by the friction loss of the throttle valve and adjusted by changing the value of the effective sectional area (A_e). As shown in (2), n means the flow coefficient (1 when inhale, –1 when exhale), b means the critical pressure ratio (0.528). P_d and P_u correspond to the downstream and upstream sides between the ventilator's output pressure and the patient's pulmonary pressure. As shown in (4) shown at the bottom of the next page, the pulmonary pressure variation generated by the patient's effort is represented as P_{mus} . The piecewise function of P_{mus} is established by fitting patient's effort curves from clinical data and refers to the description of the active lung simulator in literatures [19], [27], [28]. The amplitude of P_{mus} is represented as A . As shown in Fig. 3, the curve of P_{mus} consists of four phases, t_{fall} , t_{hold} , t_{rise} , and the expiration.

During the ventilation process, clinicians set the trigger level (P_{trig}) and the support level (P_{psv}) of the ventilator. The ventilator works when the patient's breathing effort touches P_{trig} . The blow provides the air pressure from PEEP to P_{psv} . The cycling criteria (CC) determines the cycling time from inspiration to expiration. The value of peak flow times CC equals to the cycling level of the ventilator. When the flow reaches this level, the ventilator's support pressure decreases from P_{psv} to PEEP. The simulated results compared to the clinical data are shown as Fig. 4.

By observing the ventilator's waveforms, clinicians analyse the patient-ventilator interactions and recognize PVAs. In this paper, four types of PVAs that commonly happen in clinical practice are defined, namely early cycling (EC), late cycling (LC), delayed trigger (DT), and ineffective trigger (IT). As shown in Fig. 5, the PVAs are divided into different districts using different colours. The axis $\Delta T_{expiration}$ means the time

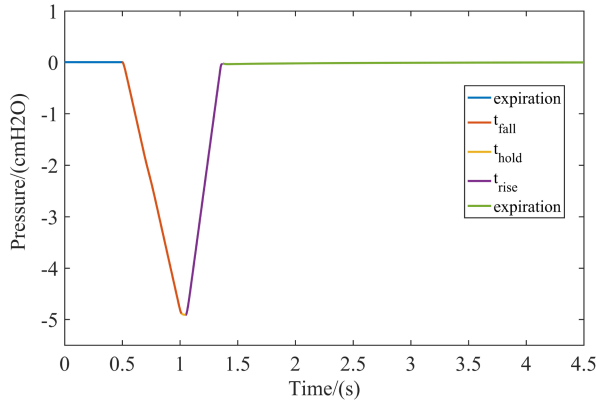


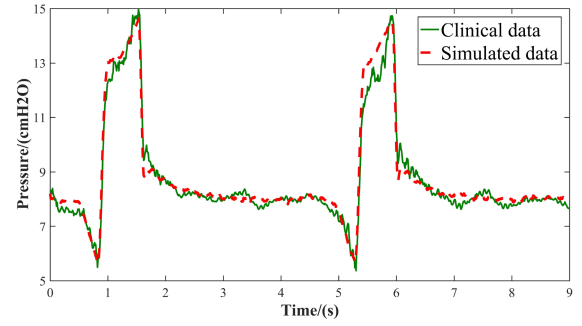
Fig. 3. The curve of P_{mus} generated from pneumatic model.

difference between the ventilator's cycling time and the start time of the patient's expiration. The axis $\Delta T_{\text{inspiration}}$ means the time difference between the ventilator's trigger time and the start time of the patient's breathing effort. The EC means that the ventilator stops the pressure support more than 150 ms before the patient's expiration. The LC means that the ventilator stops the pressure support more than 300 ms after the patient's expiration. The DT represents that the ventilator is triggered more than 200ms after the start time of the patient's breathing effort. The IT occurs when the patient's effort cannot trigger the work of ventilator.

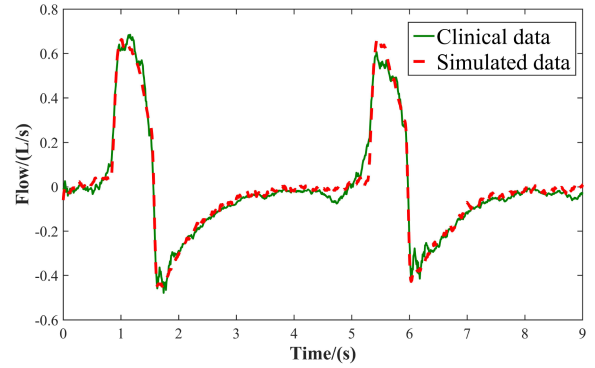
In clinical practice, it is difficult to directly obtain the start and end times of the patient's breathing effort from the mechanical ventilation system without the esophageal balloon manometry. In the simulation, the patient's breathing effort is totally controlled. Therefore, the ventilation waveforms containing different types of PVA signals can be generated by adjusting the interactions between the patient and ventilator systems. The environment of RL algorithm in this article applies this simulator to generate data at each step.

C. Algorithm Design Based on Reinforcement Learning

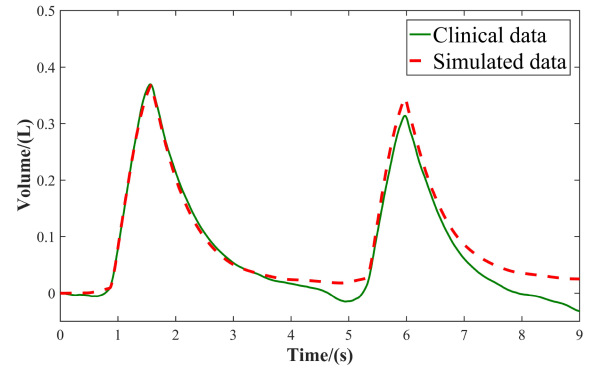
The theory and training process of RL algorithm is the most challenging part in this research. Before the construction of the network and training program, we should prepare the environment of RL. The clinical data cannot always meet the requirement of the training process. Thus, the construction of the simulated environment is essential to this algorithm. The former part describes a pneumatic model to generate the simulated breathing waveforms. During the training process, the model can generate new results for updating states and choosing actions. The PVAs occur and vary under different conditions of different patients, diseases, and ventilation stages. Considering



(a)



(b)



(c)

Fig. 4. Plotting the data generated by the pneumatic model compared with clinical data. (a) Pressure waveform; (b) flow waveform; (c) tidal volume waveform.

the complexity of data, a deep RL algorithm is used to optimize the decision-making for ventilator settings, thereby improving patient-ventilator synchrony. The ventilator is regarded as the agent in this algorithm. Different values of the agent's states mean the different patient-ventilator interactions. As shown in (5), the state tuple consists of two parameters, $\Delta T_{\text{inspiration}}$ and

$$P_{\text{mus}} = \begin{cases} -A \cdot \sin\left(\frac{\pi t}{2t_{\text{fall}}}\right), & 0 \leq t < t_{\text{fall}} \\ -A, & t_{\text{fall}} \leq t < t_{\text{fall}} + t_{\text{hold}} \\ -A \cdot \sin\left(\frac{\pi(t+t_{\text{rise}}-t_{\text{fall}}-t_{\text{hold}})}{2t_{\text{rise}}}\right), & t_{\text{fall}} + t_{\text{hold}} \leq t < t_{\text{fall}} + t_{\text{hold}} + t_{\text{rise}} \\ 0, & t_{\text{fall}} + t_{\text{hold}} + t_{\text{rise}} \leq t < T_{\text{cycle}} \end{cases} \quad (4)$$

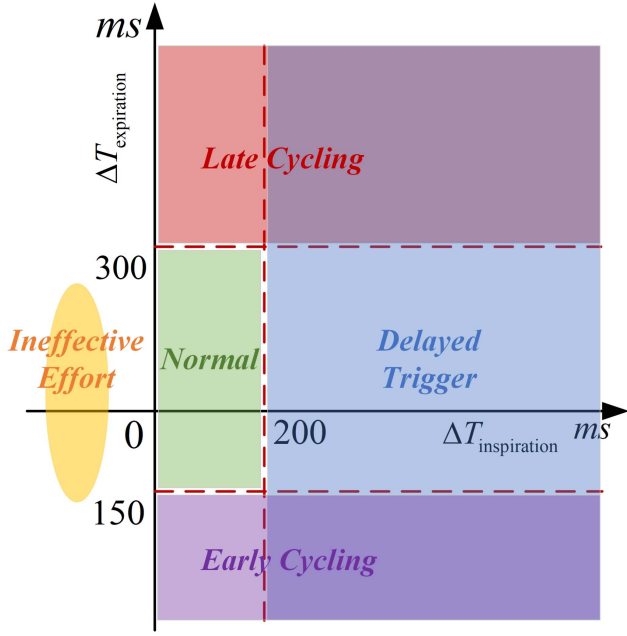


Fig. 5. Definition of the patient-ventilator asynchrony.

$\Delta T_{\text{expiration}}$.

$$S = \{\Delta T_{\text{inspiration}}, \Delta T_{\text{expiration}}\} \quad (5)$$

The actions are performed by adjusting control settings of the ventilator. In this research, P_{trig} , CC, and PEEP are considered as the actions. The action $a_t \in A$ at each time step is chosen from a finite three-dimensional set of 27 actions. The action space A is shown as (6), where $a_t[0] \in \{0, 1, 2\}$ indicates keeping, increasing, or decreasing the P_{trig} , $a_t[1] \in \{0, 1, 2\}$ indicates keeping, increasing, or decreasing the CC, and $a_t[2] \in \{0, 1, 2\}$ indicates keeping, increasing, or decreasing the PEEP.

$$A = \left\{ \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \dots, \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix} \right\} \quad (6)$$

The reward function is defined as a combination of two sigmoid functions that reward well-synchronized ventilation while penalizing PVAs, as shown in (7).

$$\begin{cases} r_{t+1} = r_{t+1}^{\text{tri}} + r_{t+1}^{\text{cycling}} \\ r_{t+1}^{\text{tri}} = \frac{1}{1+e^{-t_{\text{tri}}}} - \frac{1}{1+e^{-(t_{\text{tri}}-t_{\text{DT}})}} + \frac{1}{2} \\ r_{t+1}^{\text{cycling}} = \frac{1}{1+e^{-(t_{\text{cc}}-t_{\text{EC}})}} - \frac{1}{1+e^{-(t_{\text{cc}}-t_{\text{LC}})}} + \frac{1}{2} \end{cases} \quad (7)$$

where t_{tri} and t_{cc} represent the value of $\Delta T_{\text{inspiration}}$ and $\Delta T_{\text{expiration}}$, respectively. The values t_{DT} , t_{EC} , and t_{LC} equal to 0.2, -0.15, and 0.3, respectively, which determine the edge of the ventilation synchrony. The system aims to constrain t_{tri} to be within $(0, t_{\text{DT}})$, while constraining t_{cc} to be within $(t_{\text{EC}}, t_{\text{LC}})$. In this function, r_{t+1}^{tri} and r_{t+1}^{cycling} correspond to the trigger synchrony and cycling synchrony, respectively.

The architecture of the deep RL algorithm is developed according to the research in [26]. A technique named experience replay is applied to smooth out learning and avoid oscillations or divergence of the parameters. We set a replay memory to store the experience tuples of the agent at each step. As shown in (8),

the agent's experience of each step consists of the current state, current action, step reward, and new state.

$$\mathcal{T} = (s_t, a_t, r_t, s_{t+1}) \quad (8)$$

The minimal size of the replay memory is set to 1000. When the storage of the experience tuples reaches the minimal size, the training process starts, and the replay memory is updated by the new experience. The minibatch technique is applied to train the neural networks. Moreover, the ϵ -greedy policy is applied to choose the agent's action. The ϵ -greedy policy trades off exploration and exploitation based on a probability. We explore with probability ϵ and exploit with probability $1 - \epsilon$ at each step. The agent selects a random action a_t with probability ϵ . In other situations, the agent selects the optimal action $a_t = \max_{a \in A} \hat{Q}(s_t, a)$. An attenuation gradient is set to decrease the value of ϵ .

For each episode, the environment is reset first and the current state s_t is obtained. The action a_t is selected based on the ϵ -greedy policy. After performing the time steps in the environment, the next state s_{t+1} and the reward r_{t+1} are obtained. Then the replay memory is updated. In the training process, we set the predict model and the target model. The predict model is updated at each training step. The target model is updated by obtaining the weights from the predict model every 5 steps. The current Q-values and the target Q-values are output from the predict model and the target model, respectively. Input the next state to the target model, and then apply the maximal value of the obtained target Q-values to update the estimated Q-values. The function of the update process for Q-values is shown in (9).

$$Q_{\text{new}}(s_t, a_t) = r_{t+1} + \gamma \max_{a' \in A} Q_{\text{target}}(s_{t+1}, a') \quad (9)$$

In this research, the discount value is set to 0.99, and the MSE (Mean Squared Error) loss is applied to train the neural network. After a sufficient training process, the system can choose the optimal action for each state according to the optimized policy, thereby promoting the patient ventilator synchrony.

IV. RESULTS

In this paper, the pneumatic model of the mechanical ventilation system is used to generate simulated data for the training process of RL algorithm. To validate the effectiveness of this algorithm, an experiment is set to compare the PVAs distribution before and after applying the RL algorithm. The designed experiment is based on clinical data, which are supported by the clinical trial registration, 2022-KE-412 in Beijing Chao-Yang Hospital.

A. Train the RL Algorithm Using the Simulator

During the training process, different respiratory parameters are set to simulate different patients, diseases, and breathing conditions. Each episode represents a specific admission. The main respiratory parameters are the resistance (R_r) and the compliance (C_L) of the patient's pulmonary system. As shown in Table I, the values of these parameters are set referring to the clinical experience and the relevant literature [6], [24], [29], [30]. As shown in (4), the input muscle waveform P_{mus} is generated

TABLE I
PARAMETER SETTINGS OF THE SIMULATOR FOR GENERATING BREATHS WITH DIFFERENT PVAS

	Normal Breath	EC	LC	DT	IT
Patient parameters					
A (cmH ₂ O)	5-10	5-10	5-10	3.5-4.5	1.7-2.2
t_{fall} (s)	0.25-0.35	0.25-0.35	0.25-0.35	0.25-0.35	NA
t_{hold} (s)	0.05-0.15	0.05-0.15	0.05-0.15	0.05-0.15	NA
t_{rise} (s)	0.5-0.7	0.7-0.9	0.5-0.7	0.7-0.9	NA
Ventilator parameters					
P_{trig} (cmH ₂ O)			1-3		
CC			10%-80%		
PEEP (cmH ₂ O)			3-9		
Ventilation parameters					
R_T (cmH ₂ O/L/S)	2	4-8		7-10	10-15
C_L (L/cmH ₂ O)	0.15		0.045-0.065		>0.15
Breathing rate (bpm)			12-18		

"NA" indicates that the parameter is not defined due to insufficient patient effort in IT conditions.

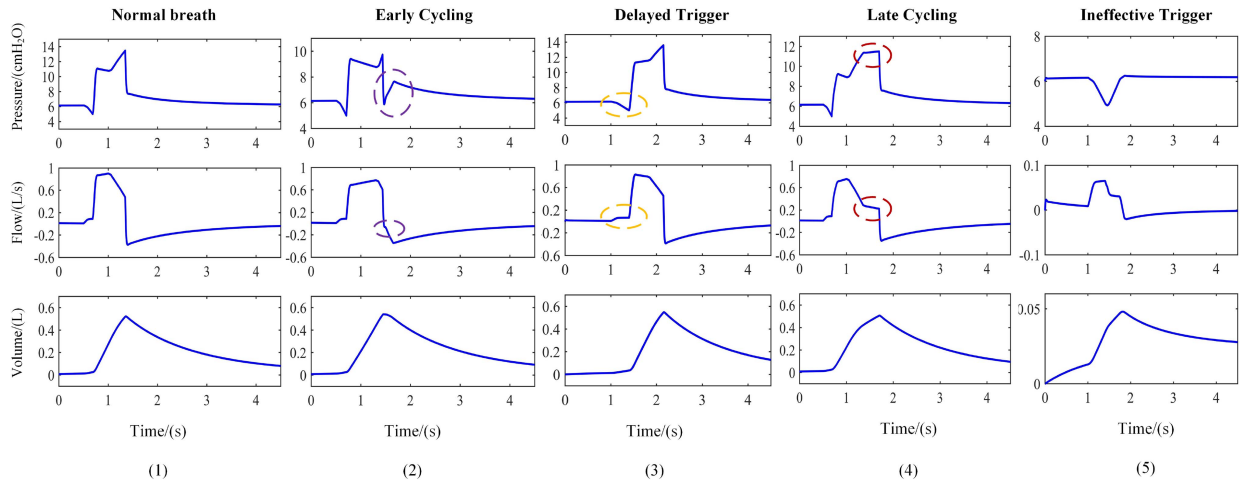


Fig. 6. The pressure, flow and volume waveforms for different types of PVA generated using the simulator.

using different parameters to simulate different PVAs. According to the description in [24], the amplitude of the muscle effort ranges from 5 to 10 cmH₂O when mimicking normal breath, EC, and LC. The amplitude ranges from 3.5 to 4.5 cmH₂O for simulating DT. The value of amplitude is 1.5–2.0 cmH₂O when IT occurs. The t_{fall} is set to 0.25–0.35 seconds. The t_{rise} is set to 0.5–0.7 seconds when mimicking normal breath and LC. The value of t_{rise} in simulated breaths with EC or DT ranges from 0.7 to 0.9 seconds.

At the beginning of each episode, the values of R_T and C_L are reset firstly to define a new admission. The values of the action and the agent's state are initialized randomly. The environment is constructed based on the simulated model, so that we know exactly the timestamps of the patient's breathing effort and can directly calculate the value of $\Delta T_{\text{inspiration}}$ and $\Delta T_{\text{expiration}}$. The ventilation parameters of the simulation are kept at the same thresholds as the ones of the clinical database. The value of P_{trig} ranges from 1 to 3 cmH₂O. The threshold of CC is 10%–80%. The PEEP level ranges from 3 to 9 cmH₂O. When designing the action space, the actions thresholds are set according to ventilation parameters. The adjustment steps of P_{trig} , CC, and PEEP are 0.5 cmH₂O, 10%, and 2 cmH₂O. In the training process, the agent's states and the Q-values are updated following the rules of RL algorithm. Besides the normal

breaths, the ventilation waveforms containing different types of PVAs are also generated, namely the EC, LC, DT, and IT, as shown in Fig. 6. The PVA features can be recognized clearly from the simulated waveforms, which are inspected visually by clinicians.

In this research, 1000 episodes are performed to cover four types of PVAs. By adjusting the respiratory parameters properly, 40 patients with different pulmonary diseases (including COPD, ARDS, and Obese) are synthesized in the simulation. At each training step, the reward value is calculated and stored. At the end of each episode, the maximal reward value, minimal reward value and average reward value of the training steps are calculated from the former stored tuples. The reward values are plotted as shown in Fig. 7. The reward values converge gradually after 600 episodes, which confirms that the training process is effective.

B. Validation of RL Algorithm Using Clinical Data

After the sufficient training of the deep Q-network, an ablation study is designed to validate the performance of this algorithm qualitatively and quantitatively. The validation experiment uses the clinical data that were collected from Beijing Chao-Yang Hospital. We select samples of breaths from these data and

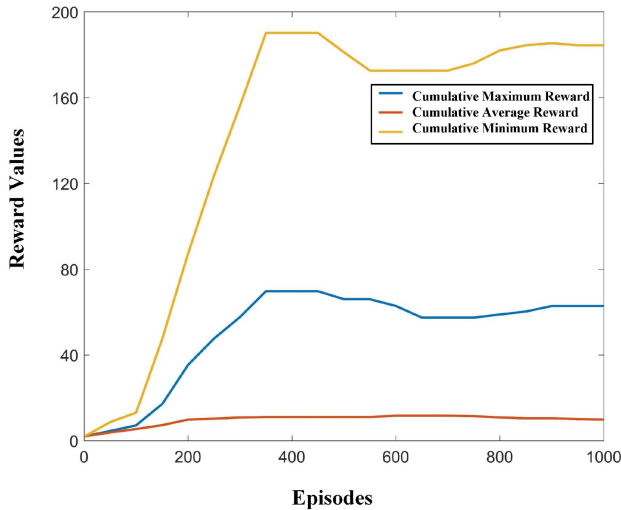


Fig. 7. Reward Values in Each Episode (Cumulative).

rebuild a clinical dataset covering all types of PVAs studied in this paper. This dataset consists of 7920 breaths under the pressure support ventilation. The sampling rate is set to 100 Hz. The values of P_{trig} , CC and PEEP are stored with the breathing waveforms. The timestamps of the patient's breathing effort are labelled by clinicians. The trigger time and cycling time of the ventilator are annotated corresponding to the breaths.

The clinical data cover the entire duration of the mechanical ventilation for each patient, including the ventilation waveforms before and after the regulation of ventilation parameters in the clinical practice. Therefore, after using the RL algorithm to adjust ventilation parameters, we can obtain the patient state values corresponding to the new ventilation parameters from the whole clinical database. To balance the distribution of the normal breaths and the breaths with PVAs, 7920 breaths of the reconstructed dataset are extracted specifically from the whole clinical database. After selecting optimized actions using RL algorithm, the state values are updated from the clinical database, but the admissions must be consistent to ensure the validity of the ablation study.

Before applying the RL algorithm, the distribution of the different types of breaths are plotted as shown in Fig. 8(a). By using the RL algorithm, we choose an optimal action for each breath and obtain a new state. The updated values of $\Delta T_{\text{inspiration}}$ and $\Delta T_{\text{expiration}}$ are determined by the original state and the selected action. According to the definition of PVAs, the classification of breaths after state update is shown in Fig. 8(b). By comparing these two figures, it can be concluded that PVAs in the clinical dataset are significantly reduced. The performance of this approach on promoting the patient-ventilator synchrony is qualitatively validated.

The performance of the RL algorithm is quantitatively analysed by counting the classification results of all breathing cycles in the clinical dataset. As shown in Table II, the breaths with PVAs account for 37.52% of all breaths in the original data. After performing the RL algorithm on the clinical data, the states of the admissions are updated. The breaths with different

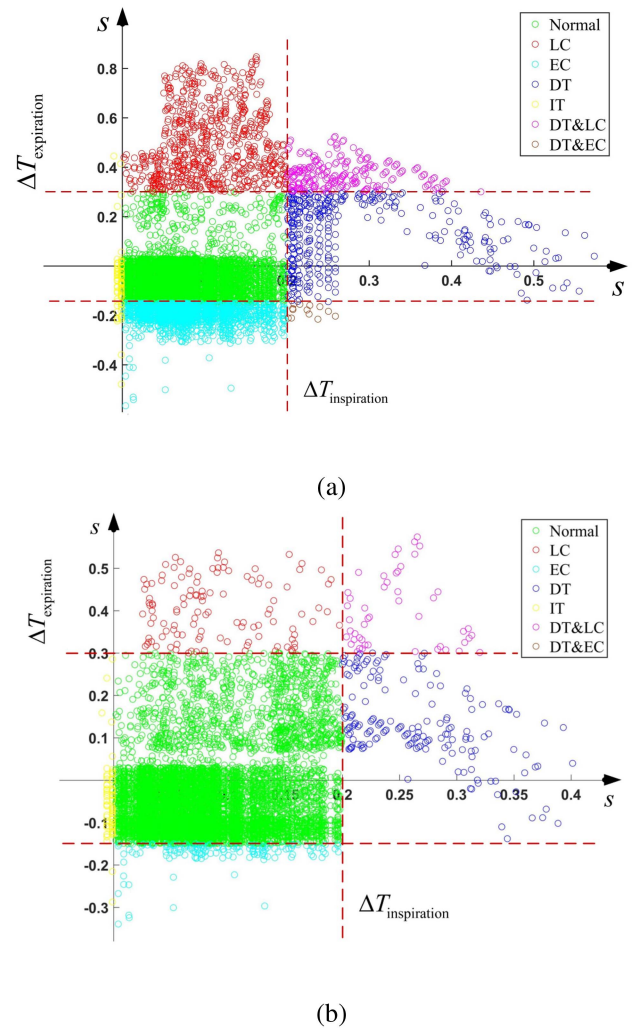


Fig. 8. The distribution of breaths containing different PVAs in clinical dataset. Each dot represents one breath cycle. Different colour means the different type of PVA. (a) shows the distribution before applying RL algorithm; (b) shows the distribution after applying RL algorithm.

TABLE II
THE CLASSIFICATION RESULTS OF BREATHS BEFORE AND AFTER USING RL ALGORITHM

	Original Data	Processed Data
Normal	4948	7359
EC	1610	151
LC	726	119
DT	334	171
IT	76	76
DT & LC	219	50
DT & EC	14	1

PVAs only account for 7.08% of all the breaths in the processed data. This approach has the most obvious effect on improving the cycling synchrony. In the original data, EC account for the highest proportion of all PVAs, reaching 20.33% of all breaths, which is followed by LC, accounting for 9.17% of all breaths. After performing RL, the proportion of EC shrinks sharply to 1.91%. The proportion of LC is reduced to 1.5%. The proportion of DT is reduced from 4.22% to 2.16%. However, this approach

cannot effectively reduce the IT phenomenon. In addition, the coupling of different types of PVAs also increase the complexity to the optimization of patient-ventilator synchrony. Through this algorithm, DT&LC and DT&EC are effectively reduced.

V. DISCUSSION

In this section, the effectiveness and clinical significance of the RL algorithm are discussed. The analysis is based on the methods and results of this study. The limitations are discussed at the end of this part.

A. Clinical Significance of the Reward Policy and Actions

The reward policy and training objectives of the RL algorithm are based on the clinical experience and medical knowledge. In clinical practice, the PVAs are recognized by clinicians observing the ventilator waveforms. Clinicians adjust the ventilator's control parameters after analysing the diagnosed types of PVAs and the patient's pathology. The trigger asynchrony (DT and IT in this study) is often associated with the patient's increasing respiratory resistance and the existence of intrinsic positive end-expiratory pressure. The value of P_{trig} , which is corresponding to $a_t[0]$ in the action space, is reduced to decrease the ventilator's trigger time. The value of PEEP, which is corresponding to $a_t[2]$ in the action space, is properly increased to overcome the patient's intrinsic positive end-expiratory pressure. These actions can improve the trigger synchrony during the ventilation. Clinicians usually lower the value of CC when EC occurs and raise the value of CC when LC occurs. The adjustment of CC is corresponding to $a_t[1]$ in the action space. The reward function established in this algorithm aims at keeping the agent's states within an ideal range. Therefore, after the sufficient training of the deep Q-network, the optimized policy can effectively learn the clinician's experience and improve the patient-ventilator synchrony.

B. Effectiveness of the Pneumatic Model for the Mechanical Ventilation System

Due to the limitation of the clinical data during training steps, the environment of RL algorithm is constructed based on a simulator of the mechanical ventilator system. In this paper, a pneumatic model is established to mimic the ventilation process. The correctness of this model has been validated in our former research [6] and [19]. As shown in Fig. 4, the simulated data can fit the clinical data well ($R^2 = 0.95$) with the same settings of the patient and ventilation parameters. The generated waveforms of the normal breath and breaths with different types of PVAs are inspected visually by clinicians from Beijing Chao-Yang Hospital and Chinese PLA General Hospital.

C. Effectiveness of the Application of RL Algorithm

This algorithm is validated qualitatively and quantitatively by performing an ablation study. The experiment is designed based on a reconstructed clinical dataset. The data were collected from patients during the treatment of the pressure support ventilation. As shown in Fig. 8, the proportion of PVAs is significantly reduced after applying RL algorithm. After performing the

optimized ventilation strategy, the breaths with PVAs decrease from 37.52% to 7.08%. Among them, the optimization is most obvious for cycling synchrony. Because the adjustment of CC can significantly affect the effect of cycling process during ventilation. The occurrence of trigger asynchrony is related to several physiological indicators of the patient, such as the respiratory resistance, intrinsic positive end-expiratory pressure, and condition changes, which makes the optimization process more complex. Nevertheless, the DT in the optimized dataset is reduced from 4.22% to 2.16%, which further demonstrates the effectiveness of this approach.

D. Limitations of This Study

The agent's state value consists of the time differences between the trigger and cycling times of the ventilator and the start and end times of the patient's breathing effort. These values are difficult to be directly obtained during ventilation without the esophageal balloon manometry. In recent years, some research on machine learning has provided methods to detect timestamps from unlabelled clinical data [25], [31], [32], [33]. These studies provide more possibilities for the clinical application of our approach. In addition, the design of actions in this algorithm is limited. We only consider the values of P_{trig} , CC, and PEEP. However, clinical strategies for promoting the patient-ventilator synchrony are more complex and require more parameters being regulated. It also results in IT not being optimized in this study.

VI. CONCLUSION

In this paper, a RL-based approach is proposed to improve the patient-ventilator synchrony during pressure support ventilation. The agent's state, action space, and reward function in this algorithm are established based on the clinical experience when decreasing PVAs. The patient admissions are simulated using the proposed pneumatic model of the mechanical ventilation system. This model constructs the environment of the RL algorithm and is used to update states during training steps. An experiment based on clinical data is designed to validate this approach qualitatively and quantitatively. After performing RL algorithm in this experiment, the proportion of the breaths with PVAs decreases significantly from 37.52% to 7.08%. This approach provides an optimized strategy for the pressure support ventilation. The trigger asynchrony and cycling asynchrony can be reduced by adjusting control parameters of the ventilator according to the optimized strategy. In the future research, the action space and reward function can be designed more complex according to the clinical experience and the patient's physiology parameters. Some specific policies can be set to deal with the IT and other types of PVAs that are not considered in this study. The sufficient tidal volume should also be considered during the optimization process.

CONFLICTS OF INTEREST

The authors declare that they have no conflict of interest.

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